INTERNET PROGRAMMING LAB

Experiment No. 5

Title: Online Flight Reservation System (SkyWays Airlines)

AIM

To design and develop an online flight reservation web application using HTML, CSS and JavaScript that lets users search flights by destination, date and class, view real-time seat availability, book and pay for tickets, print e-tickets, cancel tickets and give feedback, and lets the administrator view reports and manage flight schedules.

ALGORITHM

Step 1. Start the application and open the Login page (users: Login / Register / Forgot Password; administrator: admin login).

Step 2. Register: read username and password, validate them, check that the username is unique and store the new user.

Step 3. Login: compare the entered credentials with stored users; on success open the Search Flight page (admin is taken to the Admin Report).

Step 4. Search Flight: read From, To, Date and Class; filter the flight list for the route and display flight number, timings, fare and seats left (30 minus seats already booked for that date and class).

Step 5. Book Ticket: on clicking Book, check that the user is logged in; display the seat map with already booked seats disabled and let the user choose one seat.

Step 6. Read passenger name, age and e-mail and validate them; display the fare summary on the Ticket Payment page.

Step 7. Ticket Payment: validate the 16-digit card number, card holder and CVV; on success generate a random PNR, mark the booking "Confirmed" and store it.

Step 8. Display the e-ticket (Print Ticket page) with PNR, flight, seat and fare; show an e-mail confirmation message and allow printing.

Step 9. My Ticket / View Booking Status: list the logged-in user's bookings; for a confirmed booking the user may View/Print or Cancel it.

Step 10. Cancel Ticket: change status to "Cancelled", free the seat and show a refund of 80% of the fare.

Step 11. Customer Feedback: store rating and comment given by the user.

Step 12. Admin Report: show total, confirmed and cancelled bookings, revenue, average feedback and the list of all bookings; admin may add new flights.

Step 13. Logout and stop.

PROGRAM

File: p5.html (HTML structure, CSS styling and JavaScript logic in a single file)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>SkyWays - Online Flight Reservation</title>
<style>
:root{--c:#0b4f8a}
*{box-sizing:border-box;margin:0;padding:0;font-family:Segoe UI,Arial,sans-serif}
body{background:#f2f5fa;color:#222;line-height:1.4}
header{background:var(--c);color:#fff;padding:12px 20px;display:flex;justify-content:space-between;align-
items:center;flex-wrap:wrap;gap:8px}
nav button{background:none;border:0;color:#fff;margin-left:4px;padding:7px 11px;cursor:pointer;border-
radius:4px;font-size:14px}
nav button:hover{background:#ffffff33}
main{max-width:1000px;margin:18px auto;padding:0 12px}
.card{background:#fff;border-radius:8px;padding:16px;margin-bottom:16px;box-shadow:0 1px 4px #0002}
.page{display:none}.page.on{display:block}
h3{margin-bottom:10px;color:var(--c)}
input,select,textarea{padding:7px;margin:4px 6px 4px 0;border:1px solid #bbb;border-radius:4px;font-
size:14px}
.btn{background:var(--c);color:#fff;border:0;padding:8px 14px;border-radius:4px;cursor:pointer;font-
size:14px;margin:2px}
.btn.red{background:#b3261e}.btn.gray{background:#666}
table{width:100%;border-collapse:collapse;margin-top:8px;font-size:14px}
th,td{border:1px solid #ddd;padding:7px;text-align:left}th{background:#e9eef6}
.ok{color:#0a7a2f;margin:6px 0}.err{color:#b3261e;margin:6px 0}
.grid{display:grid;grid-template-columns:repeat(auto-fill,minmax(215px,1fr));gap:12px}
.item{border:1px solid #e1e4ea;border-radius:8px;padding:12px;background:#fff}
.pic{height:90px;border-radius:6px;display:flex;align-items:center;justify-content:center;font-
size:44px;background:linear-gradient(135deg,#dfe9f7,#f7e9df);margin-bottom:8px}
.pill{display:inline-block;background:#e5ecf8;border-radius:12px;padding:2px 10px;font-size:12px;margin:2px}
.stat{display:inline-block;min-width:150px;background:#eef3fb;border-radius:8px;padding:10px
14px;margin:4px;text-align:center}
.stat b{display:block;font-size:22px;color:var(--c)}
@media(max-width:600px){nav button{padding:5px 6px;font-size:12px}table{font-size:12px}}

.seats{display:grid;grid-template-columns:repeat(6,46px);gap:6px;margin:10px 0}
.seat{padding:8px 0;border:1px solid #888;border-radius:5px;background:#e6f4ea;cursor:pointer;font-
size:12px}
.seat.taken{background:#ccc;cursor:not-allowed;color:#777}
.seat.sel{background:#0b4f8a;color:#fff}
.ticket{border:2px dashed #0b4f8a;border-radius:10px;padding:16px;background:#f8fbff}
</style>
</head>
<body>
<header>
  <h2>&#9992; SkyWays Airlines</h2>
  <nav>
    <button onclick="go('search')">Search Flight</button>
    <button onclick="go('tickets')">My Ticket</button>
    <button onclick="go('feedback')">Feedback</button>
    <button onclick="go('admin')">Admin Report</button>
    <button id="authBtn" onclick="authClick()">Login</button>
  </nav>
</header>
<main>
  <!-- LOGIN / REGISTER / FORGOT PASSWORD -->
  <section id="login" class="page card">
    <h3>Login / Register</h3>
    <input id="lu" placeholder="Username">
    <input id="lp" type="password" placeholder="Password">
```

```
    <button class="btn" onclick="login()">Login</button>
    <button class="btn gray" onclick="reg()">Register</button>
    <a href="#" onclick="forgot();return false">Forgot password?</a>
    <p id="lmsg"></p>
    <small>Administrator demo login: admin / admin123</small>
  </section>

  <!-- SEARCH FLIGHT -->
  <section id="search" class="page card">
    <h3>Search Flight</h3>
    From <select id="from"></select>
    To <select id="to"></select>
    Date <input type="date" id="date">
    Class <select id="cls"><option value="eco">Economy</option><option
value="biz">Business</option></select>
    <button class="btn" onclick="searchFlights()">Search</button>
    <div id="res"></div>
  </section>

  <!-- BOOK TICKET -->
  <section id="book" class="page card">
    <h3>Book Ticket</h3>
    <div id="bsum"></div>
    <p>Select a seat (grey = already booked):</p>
    <div id="seats" class="seats"></div>
    <input id="pn" placeholder="Passenger name">
    <input id="pa" type="number" placeholder="Age" style="width:80px">
    <input id="pe" type="email" placeholder="E-mail">
    <button class="btn" onclick="toPay()">Proceed to Payment</button>
    <p id="bmsg"></p>
  </section>

  <!-- TICKET PAYMENT -->
  <section id="pay" class="page card">
    <h3>Ticket Payment</h3>
    <div id="psum"></div>
    <input id="cn" placeholder="Card number (16 digits)" maxlength="19">
    <input id="ch" placeholder="Card holder name">
    <input id="cv" placeholder="CVV" maxlength="3" style="width:70px">
    <button class="btn" onclick="payNow()">Pay Now</button>
    <p id="pmsg"></p>
  </section>

  <!-- PRINT TICKET -->
  <section id="print" class="page card">
    <h3>E-Ticket</h3>
    <div id="tk"></div>
    <button class="btn" onclick="window.print()">Print Ticket</button>
  </section>

  <!-- MY TICKET / CANCEL TICKET / BOOKING STATUS -->
  <section id="tickets" class="page card">
    <h3>My Tickets &amp;amp; Booking Status</h3>
    <div id="tlist"></div>
  </section>

  <!-- CUSTOMER FEEDBACK -->
  <section id="feedback" class="page card">
    <h3>Customer Feedback</h3>
    Rating <select id="fr"><option>5</option><option>4</option><option>3</option><option>2</
option><option>1</option></select>
    <input id="ft" placeholder="Your comments" style="width:50%">
    <button class="btn" onclick="sendFb()">Submit</button>
    <p id="fmsg"></p>
  </section>

  <!-- ADMIN REPORT -->
  <section id="admin" class="page card">
    <h3>Admin Report</h3>
```

```
    <div id="rep"></div>
    <h3 style="margin-top:14px">Add Flight</h3>
    <input id="nf" placeholder="Flight no" style="width:90px">
    <input id="nfrom" placeholder="From" style="width:100px"><input id="nto" placeholder="To"
style="width:100px">
    <input id="ndep" placeholder="Dep" style="width:70px"><input id="narr" placeholder="Arr"
style="width:70px">
    <input id="neco" type="number" placeholder="Eco fare" style="width:90px"><input id="nbiz" type="number"
placeholder="Biz fare" style="width:90px">
    <button class="btn" onclick="addFlight()">Add</button>
    <p id="amsg"></p>
  </section>
</main>

<script>
const $ = id => document.getElementById(id);
const load = (k, d) => { try { return JSON.parse(localStorage.getItem(k)) || d; } catch (e) { return d; } };
const save = (k, v) => { try { localStorage.setItem(k, JSON.stringify(v)); } catch (e) {} };

let users    = load('sw_users', [{u:'admin', p:'admin123', role:'admin'}]);
let bookings = load('sw_book', []);
let reviews  = load('sw_fb', []);
let flights  = load('sw_fl', [
  {id:'SW101', from:'Chennai',   to:'Delhi',     dep:'06:00', arr:'08:45', eco:4500, biz:9800},
  {id:'SW202', from:'Chennai',   to:'Delhi',     dep:'14:30', arr:'17:10', eco:5200, biz:11000},
  {id:'SW303', from:'Chennai',   to:'Mumbai',    dep:'09:15', arr:'11:10', eco:3800, biz:8200},
  {id:'SW404', from:'Bengaluru', to:'Kolkata',   dep:'18:00', arr:'20:30', eco:4900, biz:10500},
  {id:'SW505', from:'Delhi',     to:'Chennai',   dep:'07:45', arr:'10:30', eco:4700, biz:9900}
]);
let me = null, sel = null, chosenSeat = null, ctx = {};
const SEATS = [];
for (let r = 1; r <= 5; r++) for (const c of 'ABCDEF') SEATS.push(r + c);

function go(p) {
  if (['tickets', 'feedback'].includes(p) && !me) { msg('lmsg', 'Please login first.', 1); p = 'login'; }
  if (p === 'admin' && (!me || me.role !== 'admin')) { msg('lmsg', 'Administrator login required.', 1); p =
'login'; }
  document.querySelectorAll('.page').forEach(s => s.classList.remove('on'));
  $(p).classList.add('on');
  if (p === 'tickets') renderTickets();
  if (p === 'admin') renderAdmin();
}
function msg(id, t, bad) { $(id).className = bad ? 'err' : 'ok'; $(id).textContent = t; }

/* ---------- authentication ---------- */
function reg() {
  const u = $('lu').value.trim(), p = $('lp').value;
  if (!u || p.length < 4) return msg('lmsg', 'Enter username and a password of 4+ characters.', 1);
  if (users.some(x => x.u === u)) return msg('lmsg', 'Username already exists.', 1);
  users.push({u, p, role:'user'}); save('sw_users', users);
  msg('lmsg', 'Registration successful. You can login now.');
}
function login() {
  const f = users.find(x => x.u === $('lu').value.trim() && x.p === $('lp').value);
  if (!f) return msg('lmsg', 'Invalid username or password.', 1);
  me = f; $('authBtn').textContent = 'Logout (' + f.u + ')';
  msg('lmsg', 'Welcome ' + f.u + '!'); go(f.role === 'admin' ? 'admin' : 'search');
}
function forgot() {
  const f = users.find(x => x.u === $('lu').value.trim());
  msg('lmsg', f ? 'A password reset link has been sent to the registered e-mail of ' + f.u + '.' : 'Enter
your registered username first.', !f);
}
function authClick() {
  if (me) { me = null; $('authBtn').textContent = 'Login'; msg('lmsg', 'Logged out.'); }
  go('login');
}

/* ---------- search flight ---------- */
```

```
function initSearch() {
  const cities = [...new Set(flights.flatMap(f => [f.from, f.to]))].sort();
  $('from').innerHTML = $('to').innerHTML = cities.map(c => `<option>${c}</option>`).join('');
  $('from').value = 'Chennai'; $('to').value = 'Delhi';
  const d = new Date(); d.setDate(d.getDate() + 1);
  $('date').value = d.toISOString().slice(0, 10);
}
function taken(f, date, cls) {
  return bookings.filter(b => b.fid === f.id && b.date === date && b.cls === cls && b.status ===
'Confirmed').map(b => b.seat);
}
function searchFlights() {
  const from = $('from').value, to = $('to').value, date = $('date').value, cls = $('cls').value;
  const list = flights.filter(f => f.from === from && f.to === to);
  if (!list.length) { $('res').innerHTML = '<p class="err">No flights available on this route.</p>'; return;
}
  $('res').innerHTML = '<table><tr><th>Flight</th><th>Route</th><th>Dep</th><th>Arr</th><th>Fare
(Rs.)</th><th>Seats left</th><th></th></tr>' +
    list.map(f => `<tr><td>${f.id}</td><td>${f.from} &rarr; ${f.to}</td><td>${f.dep}</td><td>${f.arr}</td>
      <td>${f[cls]}</td><td>${30 - taken(f, date, cls).length}</td>
      <td><button class="btn" onclick="pick('${f.id}')">Book</button></td></tr>`).join('') + '</table>';
}

/* ---------- book ticket ---------- */
function pick(id) {
  if (!me) { msg('lmsg', 'Please login to book a ticket.', 1); return go('login'); }
  sel = flights.find(f => f.id === id); chosenSeat = null;
  ctx = {date: $('date').value, cls: $('cls').value};
  $('bsum').innerHTML = `<b>${sel.id}</b> ${sel.from} &rarr; ${sel.to} | ${ctx.date} | ${sel.dep}-${sel.arr}
| ${ctx.cls === 'eco' ? 'Economy' : 'Business'} | Rs. ${sel[ctx.cls]}`;
  drawSeats(); $('bmsg').textContent = ''; go('book');
}
function drawSeats() {
  const t = taken(sel, ctx.date, ctx.cls);
  $('seats').innerHTML = SEATS.map(s => `<div class="seat ${t.includes(s) ? 'taken' : ''} ${s === chosenSeat
? 'sel' : ''}"
    onclick="${t.includes(s) ? '' : `chooseSeat('${s}')`}">${s}</div>`).join('');
}
function chooseSeat(s) { chosenSeat = s; drawSeats(); }
function toPay() {
  const n = $('pn').value.trim(), a = +$('pa').value, e = $('pe').value.trim();
  if (!chosenSeat) return msg('bmsg', 'Please select a seat.', 1);
  if (!n || !(a > 0) || !/^\S+@\S+\.\S+$/.test(e)) return msg('bmsg', 'Enter valid passenger name, age and
e-mail.', 1);
  ctx.pax = {n, a, e};
  $('psum').innerHTML = `<p>Passenger: <b>${n}</b> | Flight: <b>${sel.id}</b> | Seat:
<b>${chosenSeat}</b></p><p>Amount payable: <b>Rs. ${sel[ctx.cls]}</b></p>`;
  $('pmsg').textContent = ''; go('pay');
}

/* ---------- payment & ticket ---------- */
function payNow() {
  const card = $('cn').value.replace(/\s/g, '');
  if (!/^\d{16}$/.test(card) || !$('ch').value.trim() || !/^\d{3}$/.test($('cv').value)) return msg('pmsg',
'Invalid card details.', 1);
  const b = {pnr:'PNR' + Math.floor(100000 + Math.random() * 900000), user:me.u, fid:sel.id, from:sel.from,
to:sel.to,
    dep:sel.dep, arr:sel.arr, date:ctx.date, cls:ctx.cls, seat:chosenSeat, name:ctx.pax.n, age:ctx.pax.a,
    email:ctx.pax.e, amt:sel[ctx.cls], status:'Confirmed'};
  bookings.push(b); save('sw_book', bookings); showTicket(b.pnr);
}
function showTicket(pnr) {
  const b = bookings.find(x => x.pnr === pnr);
  $('tk').innerHTML = `<div class="ticket"><h3>SkyWays E-Ticket &mdash; ${b.pnr}</h3>
    <p><b>${b.name}</b> (Age ${b.age})   ${b.email}</p>
    <p>Flight <b>${b.fid}</b> : ${b.from} &rarr; ${b.to}</p>
    <p>Date ${b.date} | Dep ${b.dep} | Arr ${b.arr} | Seat <b>${b.seat}</b> | ${b.cls === 'eco' ?
'Economy' : 'Business'}</p>
    <p>Fare paid: Rs. ${b.amt} | Status: <b>${b.status}</b></p>
```

```
    <p class="ok">A confirmation e-mail has been sent to ${b.email}.</p></div>`;
  go('print');
}

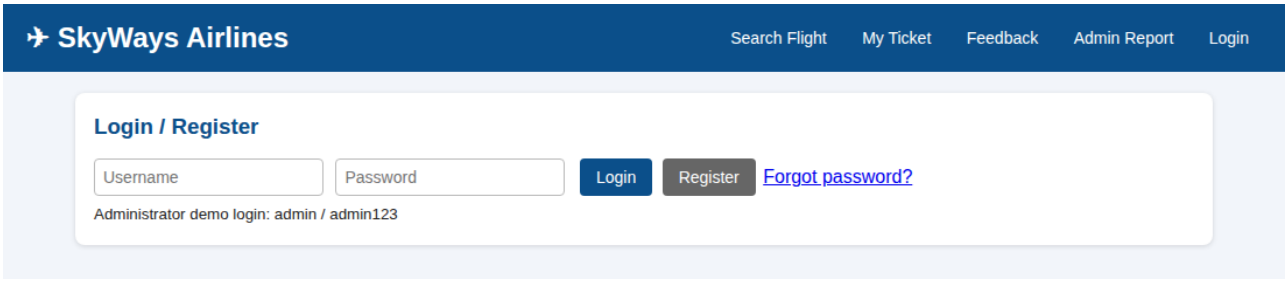
/* ---------- my tickets / cancel ---------- */
function renderTickets() {
  const mine = bookings.filter(b => b.user === me.u);
  $('tlist').innerHTML = mine.length ?
'<table><tr><th>PNR</th><th>Flight</th><th>Date</th><th>Seat</th><th>Fare</th><th>Status</th><th>Action</
th></tr>' +
    mine.map(b => `<tr><td>${b.pnr}</td><td>${b.fid} ${b.from}-${b.to}</td><td>${b.date}</td><td>$
{b.seat}</td><td>${b.amt}</td><td>${b.status}</td>
    <td><button class="btn" onclick="showTicket('${b.pnr}')">View</button>
    ${b.status === 'Confirmed' ? `<button class="btn red" onclick="cancelT('${b.pnr}')">Cancel</button>` :
''}</td></tr>`).join('') + '</table>' : '<p>No bookings yet.</p>';
}
function cancelT(pnr) {
  const b = bookings.find(x => x.pnr === pnr); b.status = 'Cancelled'; save('sw_book', bookings);
  renderTickets(); $('tlist').insertAdjacentHTML('afterbegin', `<p class="ok">Ticket ${pnr} cancelled.
Refund of Rs. ${b.amt * 0.8} (80%) will be credited.</p>`);
}

/* ---------- feedback ---------- */
function sendFb() {
  if (!$('ft').value.trim()) return msg('fmsg', 'Please write a comment.', 1);
  reviews.push({u:me.u, r:+$('fr').value, t:$('ft').value}); save('sw_fb', reviews);
  $('ft').value = ''; msg('fmsg', 'Thank you for your feedback!');
}

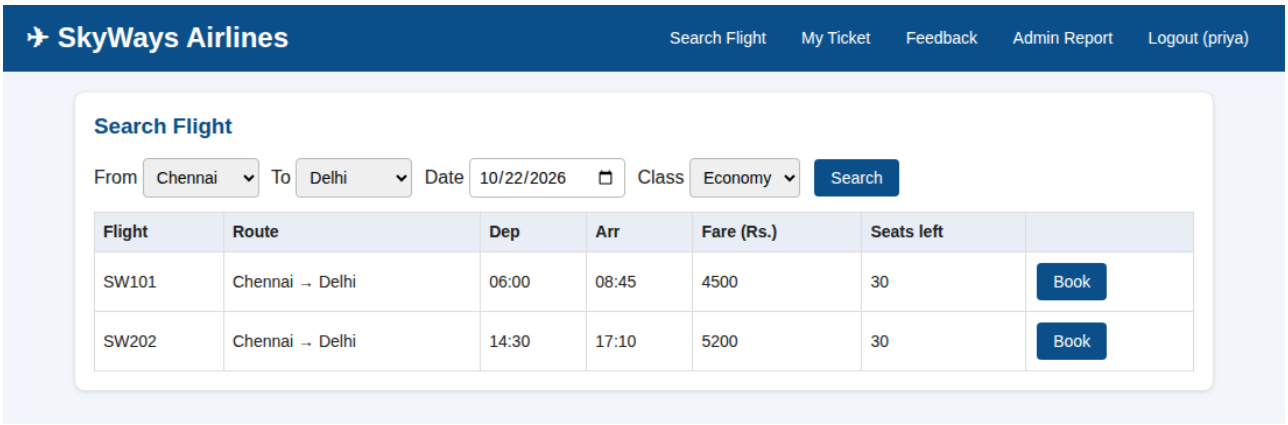
/* ---------- admin ---------- */
function renderAdmin() {
  const conf = bookings.filter(b => b.status === 'Confirmed'), canc = bookings.filter(b => b.status ===
'Cancelled');
  const rev = conf.reduce((s, b) => s + b.amt, 0) + canc.reduce((s, b) => s + b.amt * 0.2, 0);
  const avg = reviews.length ? (reviews.reduce((s, r) => s + r.r, 0) / reviews.length).toFixed(1) : '-';
  $('rep').innerHTML = `<div class="stat"><b>${bookings.length}</b>Total bookings</div><div
class="stat"><b>${conf.length}</b>Confirmed</div>
    <div class="stat"><b>${canc.length}</b>Cancelled</div><div class="stat"><b>Rs.
${rev}</b>Revenue</div><div class="stat"><b>${avg}</b>Avg. feedback</div>
    <table><tr><th>PNR</th><th>Passenger</th><th>Flight</th><th>Date</th><th>Seat</th><th>Fare</
th><th>Status</th></tr>` +
    bookings.map(b => `<tr><td>${b.pnr}</td><td>${b.name}</td><td>${b.fid}</td><td>${b.date}</td><td>$
{b.seat}</td><td>${b.amt}</td><td>${b.status}</td></tr>`).join('') + '</table>';
}
function addFlight() {
  const f = {id:$('nf').value.trim(), from:$('nfrom').value.trim(), to:$('nto').value.trim(), dep:$
('ndep').value, arr:$('narr').value, eco:+$('neco').value, biz:+$('nbiz').value};
  if (!f.id || !f.from || !f.to || !f.eco || !f.biz) return msg('amsg', 'Fill all flight fields.', 1);
  flights.push(f); save('sw_fl', flights); initSearch(); msg('amsg', 'Flight ' + f.id + ' added.');
}

initSearch(); go('login');
</script>
</body>
</html>
```

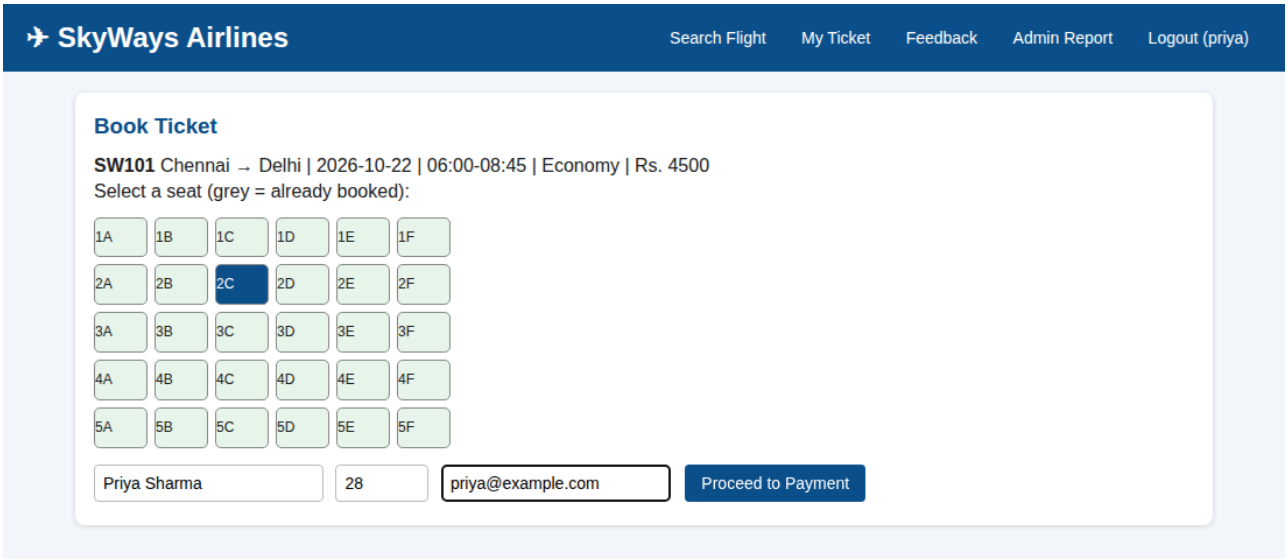
OUTPUT



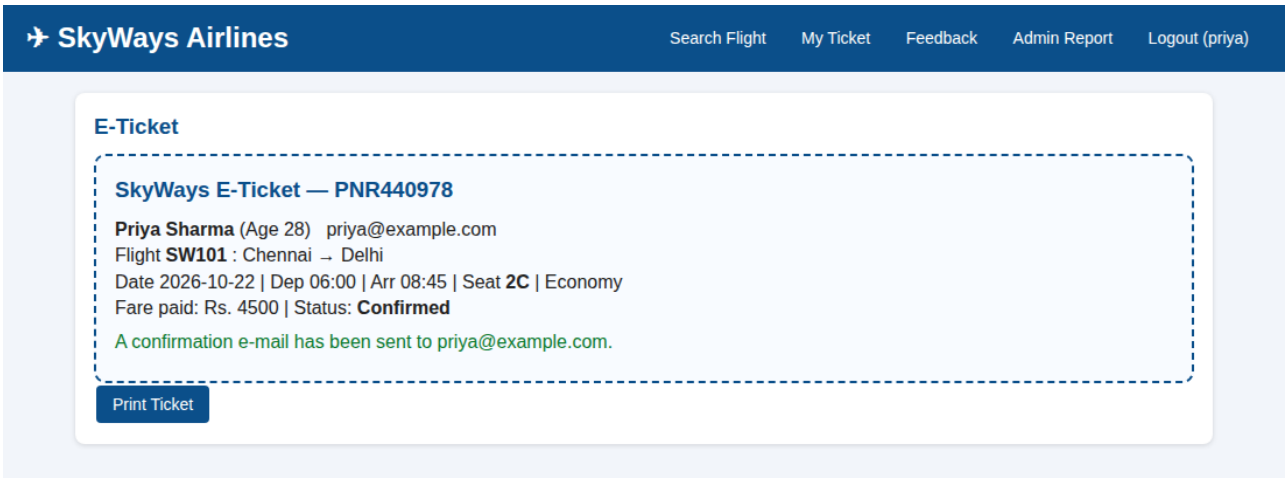
Login / Register / Forgot Password page



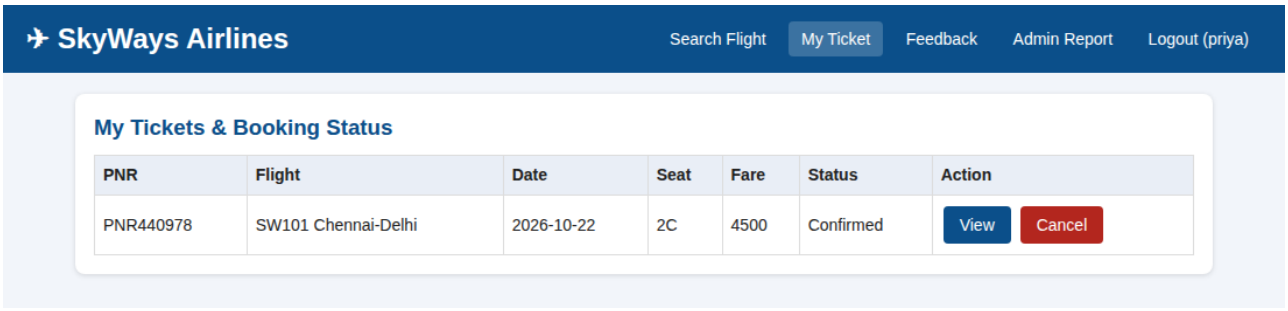
Search Flight page showing available flights and seats left



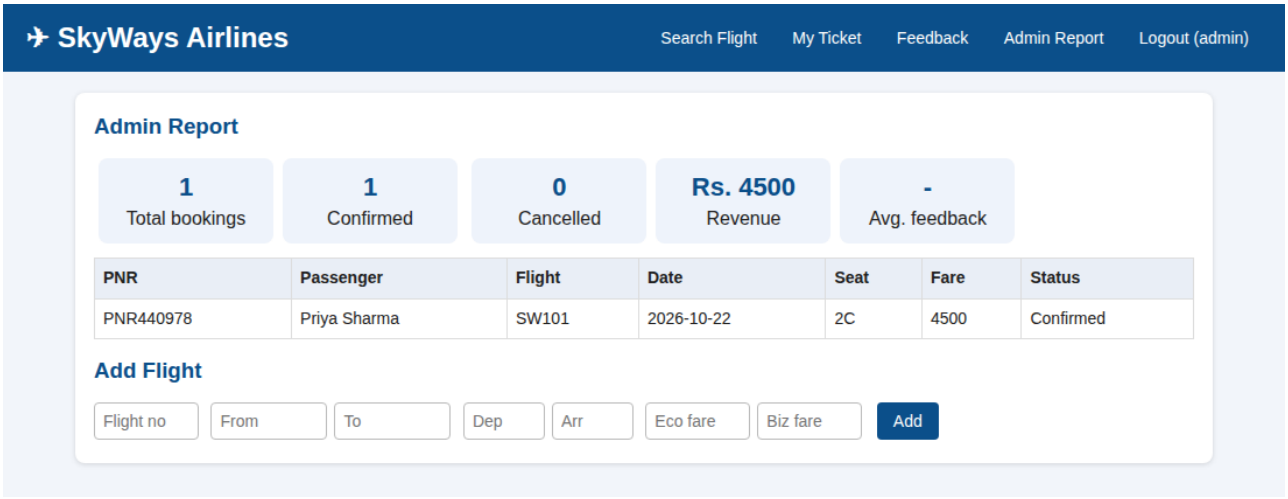
Book Ticket page - seat selection and passenger details



E-ticket generated after successful payment (PNR number)



My Ticket page - booking status with View and Cancel options



Admin Report - bookings summary, revenue and Add Flight form

RESULT

Thus the online flight reservation system was designed and implemented using HTML, CSS and JavaScript. Flights were searched, a seat was booked, payment was made, the e-ticket with PNR was generated, and the administrator report was viewed. The output was verified successfully.